

Far Western University
Faculty of science and Technology

Model Questions B Sc CSIT Entrance Examination 2077

Group A (Mathematics)

Answer all questions

Full marks: 100

Time: 2 hrs

1. Which one is not a logical statement?

(a) if you work hard, you will pass.

(c) the sun rises in the west.

(b) i am sick.

(d) 30 is a number.

2. If $|x - 2| = 5$, then x is in

(a) $\{3, 5\}$

(b) $\{2, \infty\}$

(c) $\{2, 5\}$

(d) $\{-3, 7\}$

3. $A - B$ is equal to

(a) $B - A$

(b) $\bar{A} - B$

(c) $B - \bar{A}$

(d) $B - A$

4. If $f(x) = 2x - 3$, then $f^{-1}(x)$ is equal to

(a) $(x + 3)/2$

(b) $1/(2x - 3)$

(c) $1/(3 - 2x)$

(d) $1/(2x + 3)$

5. The period of the function $f(x) = \sin(45 + 5)$ is

(a) 2π

(b) $\pi/2$

(c) π

(d) $-\pi$

6. The fourth, seventh and tenth terms of a G. P. are l, m, n respectively, then

(a) $l n = m^2$

(b) $l^2 = m^2 + n^2$

(c) $l^2 = m n$

(d) $n^2 = l m$

7. The nth term of the series $1.3 + 3.5 + 5.7 + \dots$ is

(a) $(n + 1)(n + 2)$

(b) $(n + 1)(n + 3)$

(c) $4n^2 - 1$

(d) $n(n - 1)$

8. The value of $\sin(\cos^{-1} x)$ is

(a) x

(b) $1 - x^2$

(c) $1 + x^2$

(d) $\sqrt{1 - x^2}$

9. If $\operatorname{cosec}^2 x - 2 = 0$, then x is

(a) $\pm \pi/6$

(b) $\pm \pi/4$

(c) $n x \pm \pi/4$

(d) $n x + \pi/2$

10. If $\begin{vmatrix} \lambda & 1 & 0 \\ 2 & 0 & k \\ 0 & 2 & -1 \end{vmatrix} = 0$, then k is equal to

(a) 1

(b) -1

(c) 0

(d) ± 1

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11. The value of x for which $A = \begin{vmatrix} 6 & x-2 \\ 3 & r \end{vmatrix}$ has no inverse is

- (a) 0 (b) 1 (c) -2 (d) 2

12. If the system of equation has zero solutions only then the solution is

- (a) non trivial (b) trivial (c) unique (d) particular

13. $(1+i)^6 + (1-i)^6$ is equal to

- (a) 0 (b) 1 (c) 1 (d) -1

14. $\left(\frac{2i}{1+i}\right)^2$ is equal to

- (a) -i (b) 1+i (c) 2i (d) -2i

15. If one root of $5x^2 + 2x - k = 0$ is the reciprocal of the other, then k is equal to

- (a) -5 (b) 5 (c) 2/5 (d) 5/2

16. The equation $x^2 - 2xy + y^2 + 3x - 2 = 0$ represents

- (a) hyperbola (b) parabola (c) ellipse (d) circle

17. The area of the triangle with vertices $(1, -1)$, $(-1, 1)$ and $(-1, -1)$ is

- (a) $\frac{1}{2}$ (b) -2 (c) 2 (d) 0

18. If $y = \sqrt{x + \sqrt{x + \sqrt{x + \dots}}}$ then $\frac{dy}{dx}$ is equal to

- (a) $2\sqrt{2}$ (b) $\sqrt{x^2}$ (c) $\frac{\sqrt{x}}{2y-1}$ (d) $1/(2y-1)$

19. The critical points for $f(x) = x^3 - 3x$ are

- (a) ± 3 (b) $\pm\sqrt{3}$ (c) ± 1 (d) $0, \pm\sqrt{3}$

20. The area bounded by the x -axis and the curve $y = x^3$ and ordinates of $x = 2$ and $x = 4$ is

- (a) 60 sq. units (b) 256 sq. units (c) 240 sq. units (d) 272 sq. units

21. Which of the following are direction cosines?

- a. 1,0,0 b. 2,0,0 c. 1,1,1 d. $\frac{1}{2}, \frac{1}{2}, \frac{1}{2}$

22. The conic section is said to be hyperbola if

- a. $e=1$ b. $e>1$ c. $e<1$ d. $e\geq 1$

23. Find the sum of the series $16+8+4+2+\dots$ is

- b. 22 b. 32 c. 36 d. 40

24. If w is the cube root of unity then the value of $1+w+w^2$ is

- c. 1 b. 0 c. 2 d. 3

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25. The value of $C(12,4)+C(12,5)$ is

- d. $C(13,4)$ b. $C(13,5)$ c. $C(13,6)$ d. $C(12,5)$

26. The probability of occurring the even number in the throwing a die is

- e. $1/4$ b. $1/3$ c. $1/2$ d. $1/6$

27. The identity element of any group is

- f. Unique b. not exist c. 0 d. 1

28. The middle term in the expansion of $(1+x)^{12}$ is

- g. 6^{th} b. 7^{th} c. 8^{th} d. none

29. The derivative of a^x is

- h. a^x b. 0 c. $a^x \log a$ d. $a^x / \log a$

30. The perpendiculars drawn from the vertices on their opposite sides of triangle meet at a point which is called

- i. centroid b. incenter c. circumcenter d. orthocenter

Group B (PHYSICS)

31. The number of significant figures in 1400 are

- (a) 0 (b) 1 (c) 2 (d) 4

32. A man walks 8m forwards east and 6m towards north. The magnitude of displacement is

- (a) 14m (b) 2m (c) 8m (d) $(8^2 + 6^2)^{1/2}$ m

33. If initial velocity of a projectile is doubled, the maximum range will increase by

- (a) 2 times (b) 4 times (c) 8 times (d) 16 times

34. Which of the following friction is maximum?

- (a) rolling (b) kinetic (c) rolling + kinetic (d) static

35. If the speed of rotation of earth increases, the weight of the body on earth's surface becomes

- (a) smaller (b) zero (c) greater (d) unaffected

36. A weightless rubber balloon has 100 g of water in it. Its weight in water will be

- (a) 100 g (b) 50 g (c) > 100 g (d) zero

37. The temperature of a substance increases by 270°C . This increase is equal to

- (a) 300 K (b) 0 K (c) 27 K (d) 273 K

38. Which of the following gases possess maximum rms speed at a given temperature?

- (a) H_2 (b) N_2 (c) O_2 (d) He

39. The black body emits

- (a) line spectrum (b) band spectrum (c) continuous spectrum (d) mixed spectrum

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40. The field of view is maximum for
(a) plane mirror (b) concave mirror (c) convex mirror (d) cylindrical mirror
41. Sky seen from earth appears blue because of
(a) scattering of lights (b) dispersion of lights (c) reflection (d) polarization
42. A person cannot see objects beyond 50 cm. The power of the lens to correct the vision is
(a) + 5 D (b) – 5 D (c) – 0.5 D (d) – 2 D
43. An isolated conducting sphere is given a positive charge. Its mass
(a) increases (b) decreases
(c) remains constant (d) may increase or decrease depending on material
44. Three capacitors each of capacitance $3\ \mu\text{F}$ are connected in series. The net capacitance is
(a) $1\ \mu\text{F}$ (b) $(1/3)\ \mu\text{F}$ (c) $3\ \mu\text{F}$ (d) $9\ \mu\text{F}$
45. Kirchoff's voltage law is based on the principle of conservation of
(a) charge (b) energy (c) momentum (d) mass of charges
46. A charged particle enters in a strong magnetic field. Then its kinetic energy
(a) increases (b) decreases (c) remains constant (d) increases and becomes constant
47. The self-inductance of a straight wire is
(a) zero (b) ∞ (c) negative (d) positive
48. The photoelectric is based on the law of conservation of
(a) energy (b) mass (c) momentum (d) angular momentum of photons
49. When a radioactive element emits a β -particle, the mass number of the atom
(a) increases by one (b) decreases by one (c) remains the same (d) decreases by four
50. A wave is reflected from a rigid support. The change of phase on reflection will be
(a) 0 (b) $\pi/2$ (c) $\pi/4$ (d) π
51. An airplane moving with 300km/hr in circular height which can reach the ground in 5 second it is stopped after 3 second of its fall and then again released, the total time taken by the stone to reach the ground will be.
a. zero b. 300km/hr c. 900km/hr d. 600km/hr
52. For a given material, the young's modulus is 2 times that of rigidity modulus, its poisson ratio is
b. 1 b. 0 c. 0.2 d. 0.5
53. The viscosity of..... increase with increase in temperature.
c. Liquid increases b. Gas increase c. Liquid decrease d. Liquid and gas
54. All gas at same temperature has same

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- d. Kinetic energy b. Density c. RMS speed d. None of above
55. As pressure increase boiling point of liquid
e. Increase b. Decrease c. Remains same d. First increase than decrease
56. Zener diode is used for
f. Rectifier b. Amplification c. Stabilization d. All of above
57. The appropriate percent of a radioactive substance remained after 4 half life period is
g. 3.5 b. 6.25 c. 9.25 d. 12. 2
58. x-rays are
h. Electromagnetic wave b. Highly energetic c. Absorbed by metal d. All of Above
59. magnetic permeability of diamagnetic substance is
i. Positive b. Negative c. positive and very high d. zero
60. The ratio of specific charge of proton to that alpha particle is
j. 1:2 b. 2:1 c. 1:1 d. 4 : 1

Group C (ENGLISH)

61. It costs Rs. 100.....kilo.
a. a b. an c. The d. no article
62. Write your name.....the top of the page.
a. in b. on c. over d. at
63. Be regular in the school,
a. can you b. won't you c. would you d. at
64. Either you or Itaking this responsibility.
a. Is b. are c. were d. am
65. The indirect speech of : David says to us, " I am very busy. "is.....
a. David tells us that he is very busy. b. david tells us that he was very busy.
c. david tells us that he had been very busy. d. david tells us that he had been very busy.
66. Change the voice of following: don't throw rubbish on the road.
a. let the rubbish not be thrown on the ground. b. You are supposed not to throw rubbish on the ground.
c. rubbish should not be thrown on the ground. d. All of the given option.
67. He gave the order to leave the shipit was sinking
a. because of b. because c. therefore d. since
68. I don't knowrara lake is located.
a. where b. which c. were d. when

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69. He cango.....remain in the class.

- a. neither/nor b. either/or c. whether/or d. but/so

70. If you heat iron, it.....

- a. will expend b. expends c. would expends d. would have expends

71. I will get a letter

- a. wrote b. write c. written d. to write

72. She has her food....at home.

- a. cooked b. cooks c. to cook d. to be cooked

73. The synonym of 'uniqueness's'.....

- a. inclusive b. match c. commonness d. individuality

74. The antonym of 'emphatically' is.....

- a. forcefully b. easily c. indefinitely d. consistently

75. The words' development' takes the stress onSyllable.

- a. First b. second c. third d. fourth

76. She has been learning Japanese.....2008.

- a. For b. in c. since d. by

77. One of the boysnot done his homework today.

- a. has b. have c. had d. did

78. Has I been to pokhara, i..... The phewalake .

- a. would see b. will have seen c. could see d. would have seen

79. Either Varsa or you.....solved the paper.

- a. have b. had c. has d. were

80. The injured boy died while he.....

- a. was being treated b. was treated c. is treated d. were being treated

Group C (Computer Science)

81. The AND gate produce output one (1) when inputs are

- a. 1 and 1 b. 0 and 1 c. 0 and 0 d. 1 and 2

82. The term ' computer' is derived from

- a. Latin b. German c. French d. Arabic

83. Who is the father of computer?

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- a. Allen Turing b. Charles Babbage c. Simurcary d. AgustaAdming

84. Which device is required for the internet connection ?

- a. Joystick b. modem c. CD drive d. NIC Card

85. ALU is.....

- a. Arithmetic Logic Unit b. Array Logic Unit c. Application Logic Unit d. None of above

86. Wan stands for ...

- a. wap area network b. Wide area network c. Wide array net d. Wireless area network

87. MICR Stand for

- a. Magnetic ink Character Reader b. Magnetic ink code reader
c. Magnetic ink case reader d. None

88. Junk e-mails is also called.....

- a. spam b. spoof c. sniffer script d. spool

89. A computer cannot 'boot', If it does not have the

- a. Compiler b. Loader c. Operating System d. Assembler

90. First page of website is term as-

- a. Homepage b. Index c. JAVA script d. Bookmark

91. DOS stands for

- a. Disk Operating System b. Disk operating session c. Digital operating system d. Digital Open system

92. Who is the founder of Microsoft?

- a. Babbage b. Bill Gates c. Bill Clinton d. none of these

93. 1 Byte=?

- a. 8 bits b. 4 bit m c. 2 bit d. 9bit

94. An area of a computer that temporarily holds data waiting to be processed is

- a. CPU b. Memory c. Storage d. File

95.is the key we use to run the selected command.

- a. SHIFT b. TAB c. ENTER d. CTRL

96. RAM is an example of

- a. secondary memory b. primary memory c. Main memory d. Both (b) and (c)

97. To access properties of an object, the mouse technique to use is-

- a. Dragging b. dropping c. right-clicking d. Shift-Clicking

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98. The process of transferring files from a computer on the internet to your computer is called

- a. Downloading b. uploading c. FTP d. JPEG

99. Which of the following is not a binary number ?

- a. 001 b. 101 c. 202 d. 110

100. The first computer used in Nepal was

- a. IBM 1401 b. IBM 1402 c. IBM 1403 d. CDC 3600